

LUIS FERNANDES

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TECHNICAL SKILLS

Embedded Systems	FPGA (VHDL/Verilog) • Vivado • SoC Design • ESP32 • UART • SPI • I2C • RTOS • LoRa
Systems & HPC	Linux/Unix (10+ years) • C • C++20 • x86-64 ASM • CUDA • OpenCL • OpenMP • MPI • TCP/UDP
Scripting & Infra	Rust • Python • Bash • CMake • Ninja • GTest • Clang-Tidy • Git • CI/CD • Doxygen • Docker • Nginx
Analysis & Graphics	Valgrind • GDB • Perf • NVIDIA Nsight • MATLAB • Deep Learning • Scikit-learn • NumPy • Qt • SDL2
Languages	French (Native) • English (Full Professional)

EDUCATION

California State University (CSULB), Long Beach

Aug 2025 – May 2026

Study Abroad at the Beach Program

- **Relevant courses:** Computer Logic Systems, SoC Design, CS Concepts, DSP, Machine Learning, Data Analysis.

Epitech — European Institute of Technology (5 years), Paris

Expected June 2027

Master's Degree in Systems & Software Engineering

- **Low-level:** C/x86-64 ASM Libc (printf, realloc, GNL), Linux binary tools (strace, nm, objdump) via ptrace/ELF, C++ SPICE engine.
- **Systems:** GPGPU Chess Engine (CUDA), LSTM Predictor, UDP-based ECS, Huffman Codecs, Real-time ATC Simulation, Signals/IPC

SELECTED AEROSPACE PROJECTS

LBR Telemetry & Ground Control • NASA Student Launch (USLI) • Long Beach Rocketry

Spring 2026

- Architected a low-latency C++/LoRa telemetry pipeline featuring an IPC bridge to a Qt-based Ground Control Station.
- Integrated Doppler compensation for high-velocity tracking; achieved **<20ms end-to-end latency** for mission-critical Flight Software (FSW) data streams through a Semtech SX1262 board for trajectory profiling.

BLITZAR • Astrophysical HPC Simulation • youtu.be/j9Kk4c3_0WE

Feb 2026 – Present

- Built an open-source N-Body/SPH engine in CUDA using Octree structures for **100M particles**, destined for scientific research.
- Optimized for galaxy simulation via SoA, L2 cache, and memory coalescing; achieved **92% VRAM bandwidth** on RTX 4070.
- Ensured fidelity and stability with Leapfrog/RK4 integrators, and met **NASA standards** (NPR-7150.2D, NASA-STD-8739.8B).

Silicium Network • Capstone Project (Founder, Team Lead)

2023 - Present

- Leading a 3-member team building a Solana-based (Rust/Anchor) decentralized compute grid for low-power IoT devices.
- Architected recursive delegation protocols with a **<128 KB** memory footprint for autonomous and safe edge verification.
- Developed a custom dockerized C++ binary ELF partitioner optimized for x86-64/AArch, **reducing task overhead by 30%**.
- Integrated SPL-tokens for incentive economic logic and decentralized governance for real-time worker load management.

Relativistic Raytracer • Parallel Computing • youtu.be/TMcS_57sPfY

2023

- Integrated geodesic equations using numerical integration to model gravitational lensing effects under General Relativity.
- Accurately modeled Schwarzschild and Kerr metrics using CUDA four-vector matrices; Achieved **140+ frame/hour** in 8K export.

PROFESSIONAL EXPERIENCE

Software & Systems Engineering Intern • X'PROCHEM (R&D Pharma) • Lille, France

Apr – Aug 2025

- Deployed a protein synthesis laboratory software (LIMS) to automate stages like SPPS, Cleavage, Splitting, and Analysis.
- **Reduced search latency by 40%** by using atom-based PostgreSQL queries for research workflows on shared projects and data.
- Built an auto-split tool suggesting optimal synthesis configurations to minimize manual chemist workload and resources.
- Automated analytical metrics like pI, isotopes, and solubility based on subatomic properties and **reduced false positives by 28%** in developing a **high-precision (0.0001 Da) anomaly detection** system for multicharge mass spectrometry.

QA Tester & QA Android Developer • Monstock (Supply Chain) • Reims, France

Oct 2023 – Feb 2025

- Reduced release-blocking **defects by 38%** through documenting **100+ critical bugs** and writing automated functional test suites.
- Engineered a Java/Appium pipeline, **cutting regression testing cycles by 55%** and ensuring a stable software ecosystem.

LEADERSHIP & COMMUNITY

Technical Workshop Lead • Epitech Electronics Club • Lille, France

Fall 2024

- Led weekly logic **workshops for 15 students**, using Logisim simulations and hardware prototyping to bridge theory.
- Designed **10+ modular projects** such as flip-flops, hardware-level memory, and multi-stage transistor amplifiers.

ILArD Hacklab • hacklab.fr • Charleville-Mézières, France

2014 – Present

- Member of an open-source community active in hardware hacking, install parties, on-board prototyping, and 3D printing.
- Contributed to community initiatives, including hackathons, tech fairs, and remote meetings via IRC channels.